

Spectral statistics of spike trains across three CRCNS datasets

Exploratory cross-dataset analysis, 2026-06 · fresh re-derivation, firing-rate + burst controlled, bootstrap 95% CIs

What we measure

For each unit we characterise the inter-spike-interval (ISI) spacing distribution against random-matrix spectral classes. `ks_gue` = KS distance of the unit-mean-normalised ISI spacings from the GUE Wigner surmise: LOW = rigid / level-repulsive (sub-Poisson), HIGH = far from GUE (Poisson or clustered). CV2 (Holt 1996) is a rate-robust local-irregularity measure: ~1 Poisson, <1 regular, >1 bursty/clustered.

Selectivity ↔ spectral class (does functional tuning predict ISI structure?)

Dataset (preparation)	Selectivity test	Spearman ρ (controlled)
macaque V1 – pvc-11 (Kohn)	OSI ↔ <code>ks_gue</code>	+0.72 (n=210, rate-ctrl)
rat hippocampus – hc-3 (Buzsáki)	spatial-info ↔ <code>ks_gue</code>	+0.37 [0.29, 0.46] rate-strat.
mouse retina – ret-1	RF-SNR ↔ <code>ks_gue</code>	+0.05 null (axis-mismatch)
mouse V1 – Allen (non-CRCNS ref.)	OSI ↔ <code>ks_gue</code>	-0.22 (sign reversal)

Positive = more selective cells have ISI spacings farther from GUE (more clustered). All values control firing rate and burst fraction; hc-3 also rate-stratified within quartiles (raw +0.69 → partial +0.47 → stratified +0.37) and is rate-modulated (stronger at high rate). Macaque V1 and rat hippocampus AGREE in direction; the lone reversal is mouse V1.

Local irregularity (CV2): a sensory → hippocampal clustering gradient

Hippocampus is genuinely fast-clustering with a within-structure gradient CA3 1.25 > EC 1.10 > DG 1.02 (recurrent CA3 burst-prone → sparse dentate near Poisson). Retina CV2 ≈ 1.08; mouse V1 ≈ 1.0 (Poisson point). CV2 (vs the global ISI CV) separates genuine bursting from slow rate-drift / recording-epoch structure.

Headline

In both CRCNS datasets with a matched tuning axis — macaque V1 (orientation) and rat hippocampus (spatial information) — functional selectivity predicts spectral class in the SAME direction: more selective cells are more temporally clustered (farther from GUE), robust to firing-rate and burst control. The one sign reversal is mouse V1, suggesting a species/preparation-systematic effect rather than a cortex-vs-hippocampus one.

Caveats

- `ks_gue` is rate-sensitive at finite spike counts — all correlations are rate-controlled; naive uncontrolled values are larger and should not be used.
- Burst fraction co-varies with `ks_gue` by construction (intrinsic); the reported links are the EXTRINSIC selectivity effect AFTER burst control — not merely 'place/selective cells burst.'
- DG is underpowered (n=16).
- Exploratory analyses of public CRCNS data, not peer-reviewed.
- Datasets: CRCNS pvc-11 (Kohn), hc-3 (Mizuseki/Buzsáki), ret-1; reference mouse-V1 = Allen Neuropixels. Reproducible scripts + per-cell tables available on request. (Confirm exact dataset citations before publication.)